

Project Design Phase-II

Data Flow Diagram & User Stories

Date	14 July 2026
Team ID	
Project Name	
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Level 0 (Context Diagram)

Data Flows:

- Customer → System: registration/login info, product search/filter requests, cart actions, checkout/order details
- System → Customer: product listings, cart summary, order confirmation, order status
- Admin → System: product add/update/delete requests, order status updates
- System → Admin: product catalog data, order list, sales data

Level 1 (Major Processes)

1. Authentication (1.0) — Customer/Admin credentials ↔ Firebase Auth ↔ User session
2. Product Catalog Management (2.0) — Admin adds/edits/deletes products ↔ Product DB (MongoDB) → Customer views/searches/filters products
3. Cart Management (3.0) — Customer add/update/remove items ↔ Cart state (Context API) → Cart summary
4. Checkout & Order Processing (4.0) — Customer submits address + cart ↔ Order DB (MongoDB) → Order confirmation
5. Order Tracking (5.0) — Customer/Admin views/updates order status ↔ Order DB → Order history/delivery status

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Registration	USN-1	As a customer, I can register using my email and password	I can create an account and access my dashboard	High	Sprint-1
Customer	Login	USN-2	As a customer, I can log in with my registered email and password	I can access my account after valid login	High	Sprint-1
Customer	Product Browsing	USN-3	As a customer, I can view all available products with categories	I can see product name, price, image, and stock status	High	Sprint-1
Customer	Product Search & Filter	USN-4	As a customer, I can search and filter products by category	Search/filter returns matching products only	Medium	Sprint-1
Customer	Cart Management	USN-5	As a customer, I can add products to my cart and adjust quantity	Cart updates item count and total price correctly	High	Sprint-2
Customer	Cart Management	USN-6	As a customer, I can remove items from my cart	Removed item disappears and total recalculates	Medium	Sprint-2
Customer	Checkout	USN-7	As a customer, I can enter my address and place an order	Order is created with correct items, address, and total	High	Sprint-2
Customer	Order Tracking	USN-8	As a customer, I can view my past orders and delivery status	Order history shows status: Processing/Shipped/Delivered	Medium	Sprint-3
Admin	Admin Login	USN-9	As an admin, I can log in with my authorized email	Only whitelisted admin email can access the admin panel	High	Sprint-2